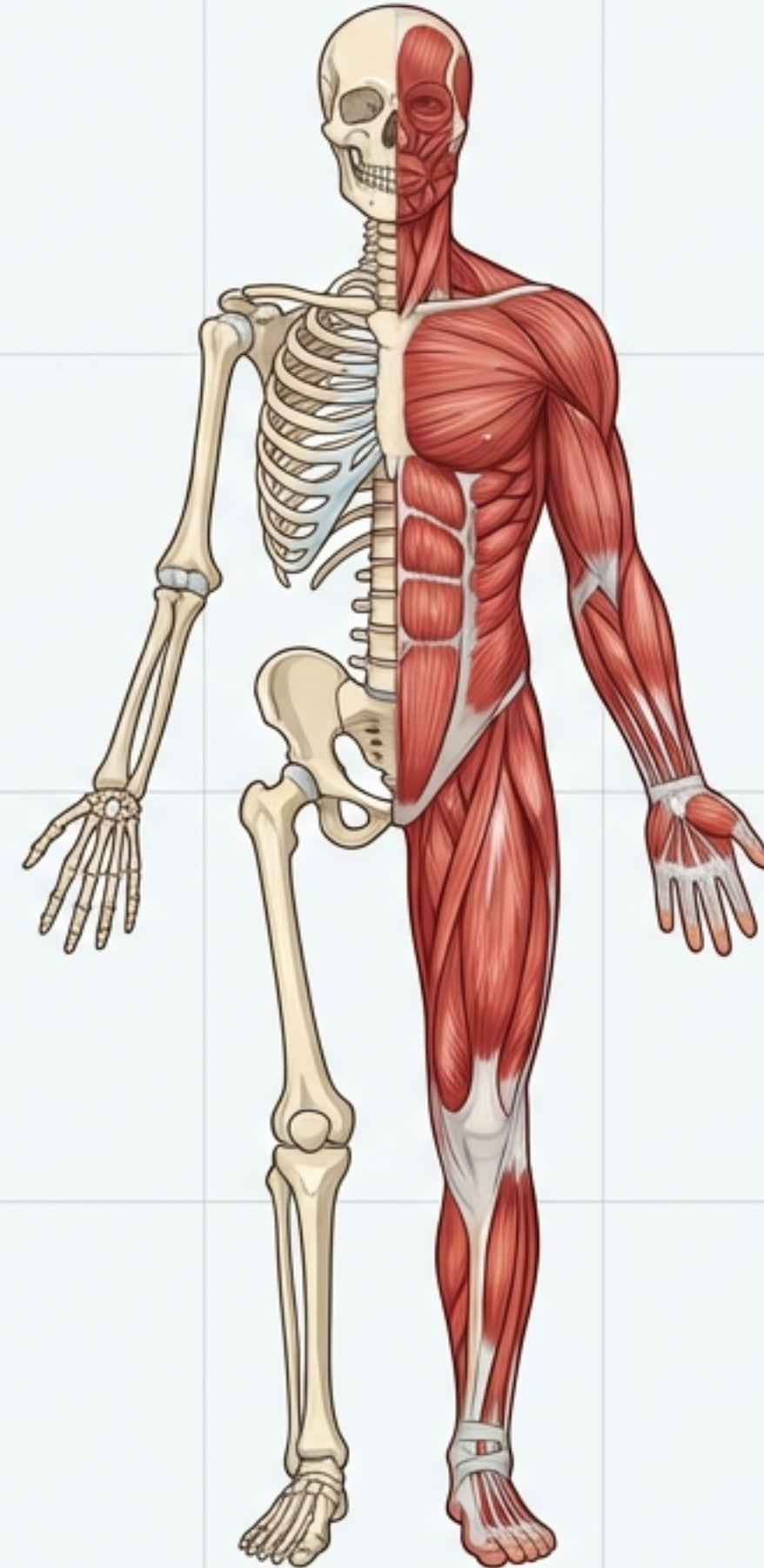


The Musculoskeletal System: A Biological Machine

Structure, Function, and Mechanics

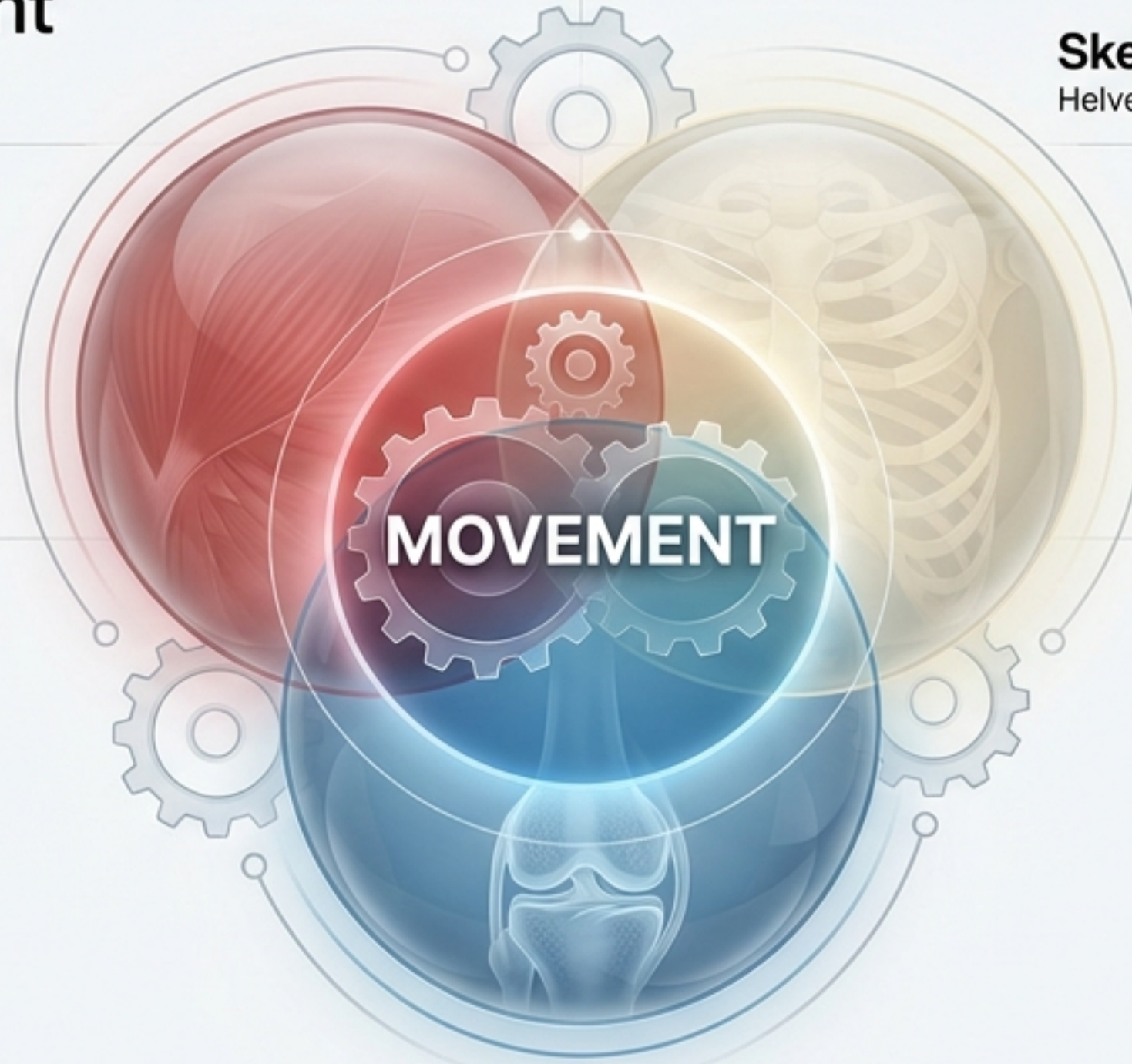
An anatomical overview of the structures that enable human movement.

Core Concept: Movement is the result of a triad of systems working in concert: the rigid framework (Skeleton), the active force generators (Muscles), and the articulation points (Joints).



⊕ The Triad of Human Movement

Muscles
Helvetica Now Display



Joints
Helvetica Now Display

Skeleton
Helvetica Now Display



Skeleton: The structural framework.



Muscles: The engine that provides active force.



Joints: The pivot points that allow articulation.

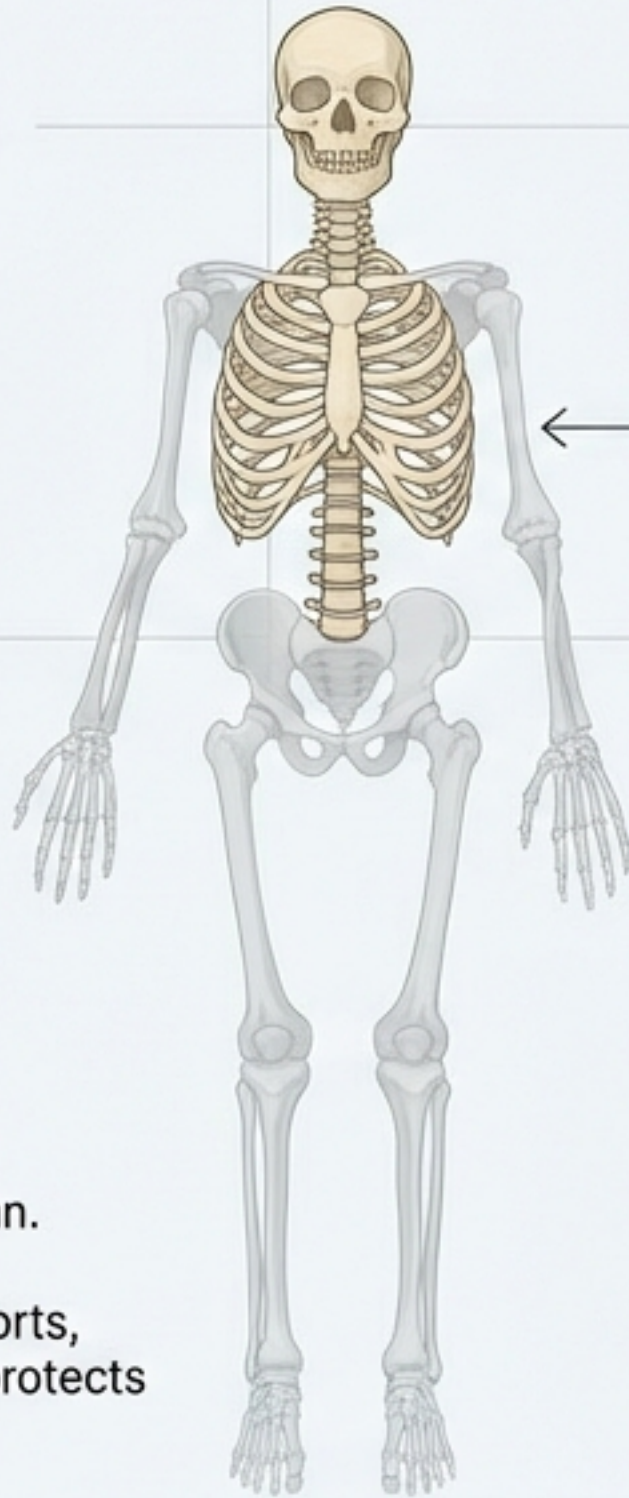


Intersection: When these three function together, they create the discipline of Kinesiology.



The Chassis: Axial and Appendicular Divisions

AXIAL SKELETON

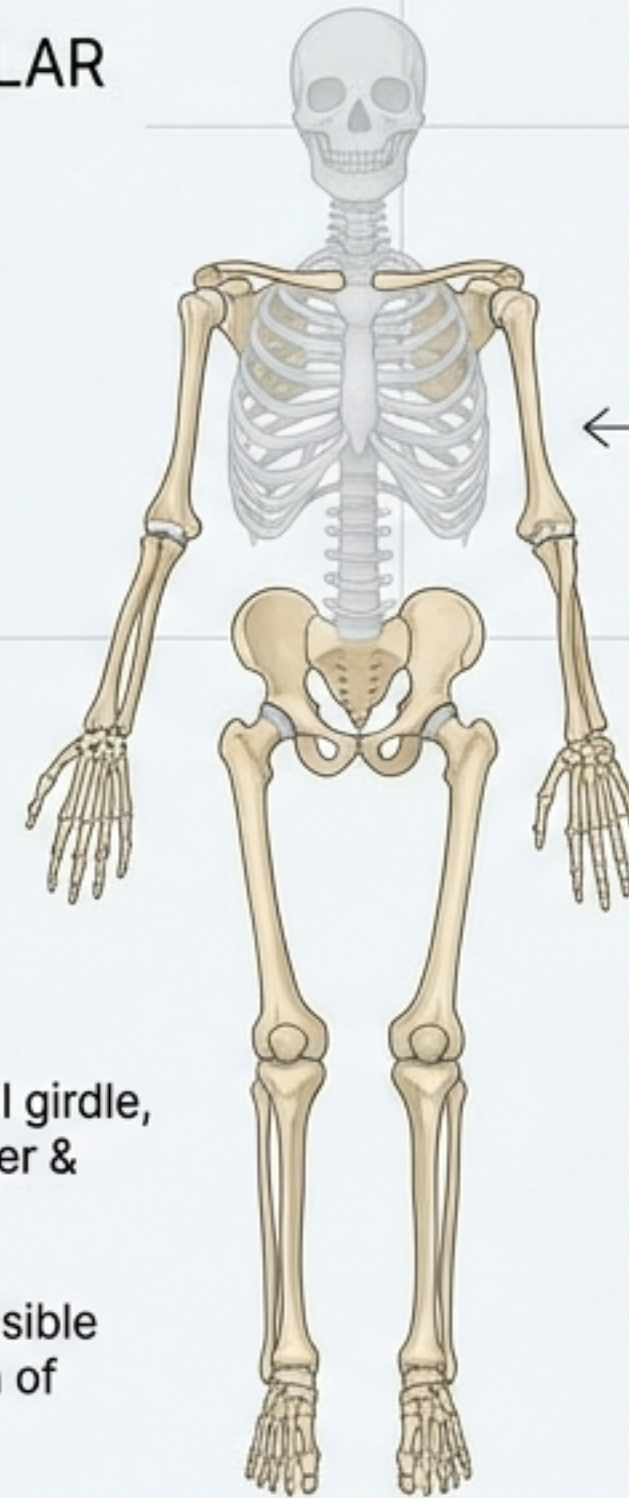


← **Axial Skeleton**
(80 Bones)

Includes: Skull, Sternum, Ribs, Vertebral Column.

Function: Supports, stabilizes, and protects vital organs.

APPENDICULAR SKELETON



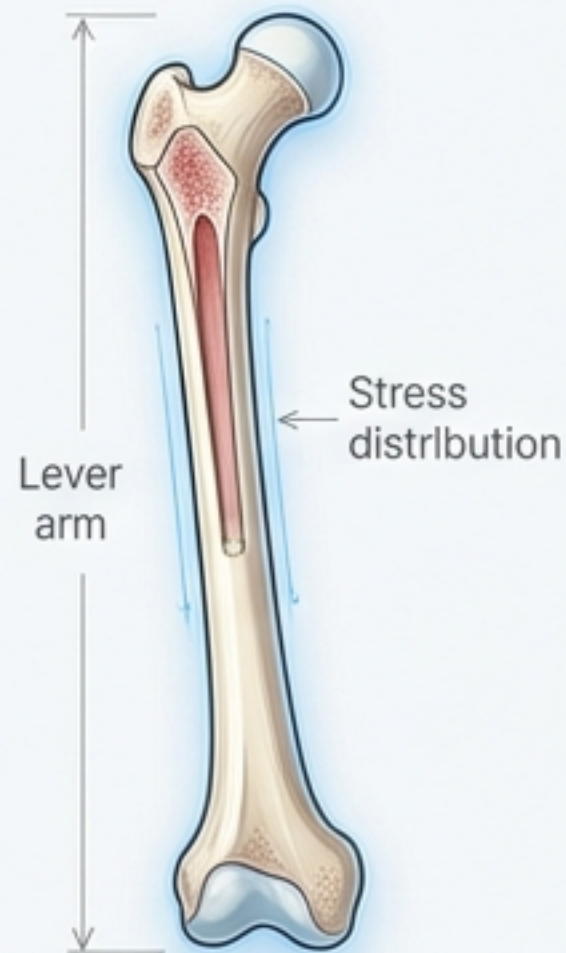
← **Appendicular Skeleton**
(126 Bones)

Includes: Pectoral girdle, Pelvic girdle, Upper & Lower limbs.

Function: Responsible for a large portion of movement.

Form Follows Function: Bone Classifications

Long Bones (Levers)



Facilitate large movements.

Short Bones (Shock Absorbers)



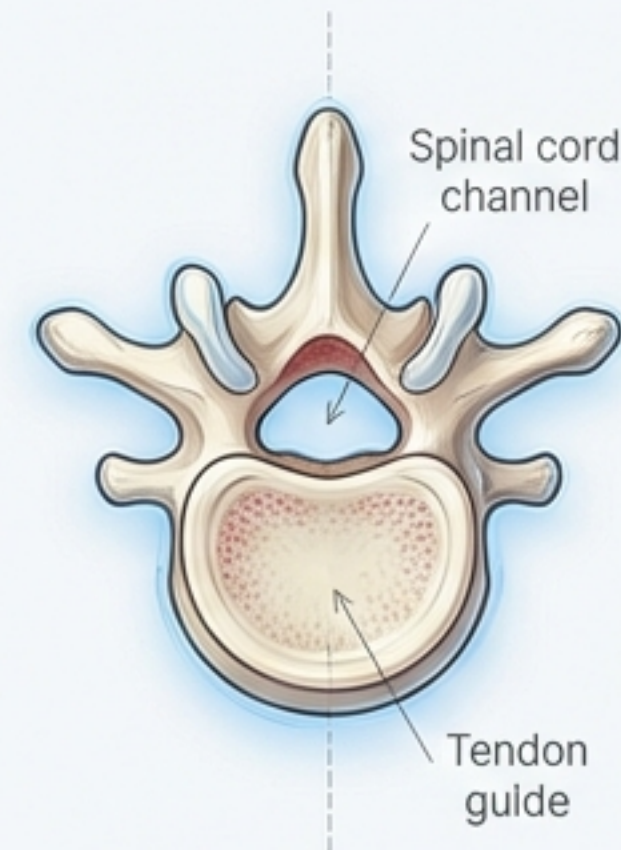
Strength and compactness.

Flat Bones (Protection)



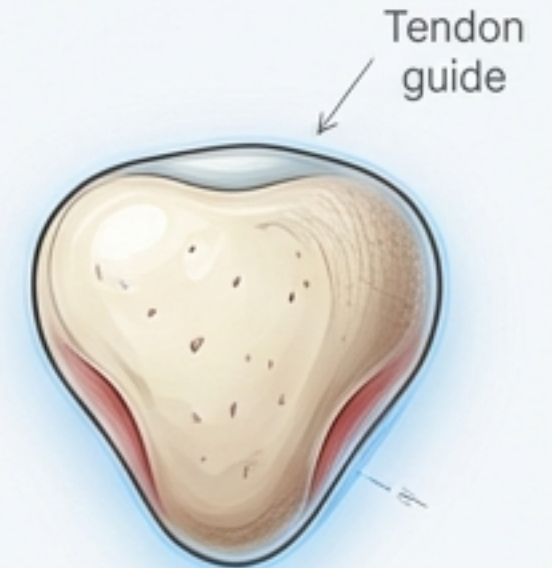
Broad surfaces to shield organs.

Irregular Bones (Special Function)



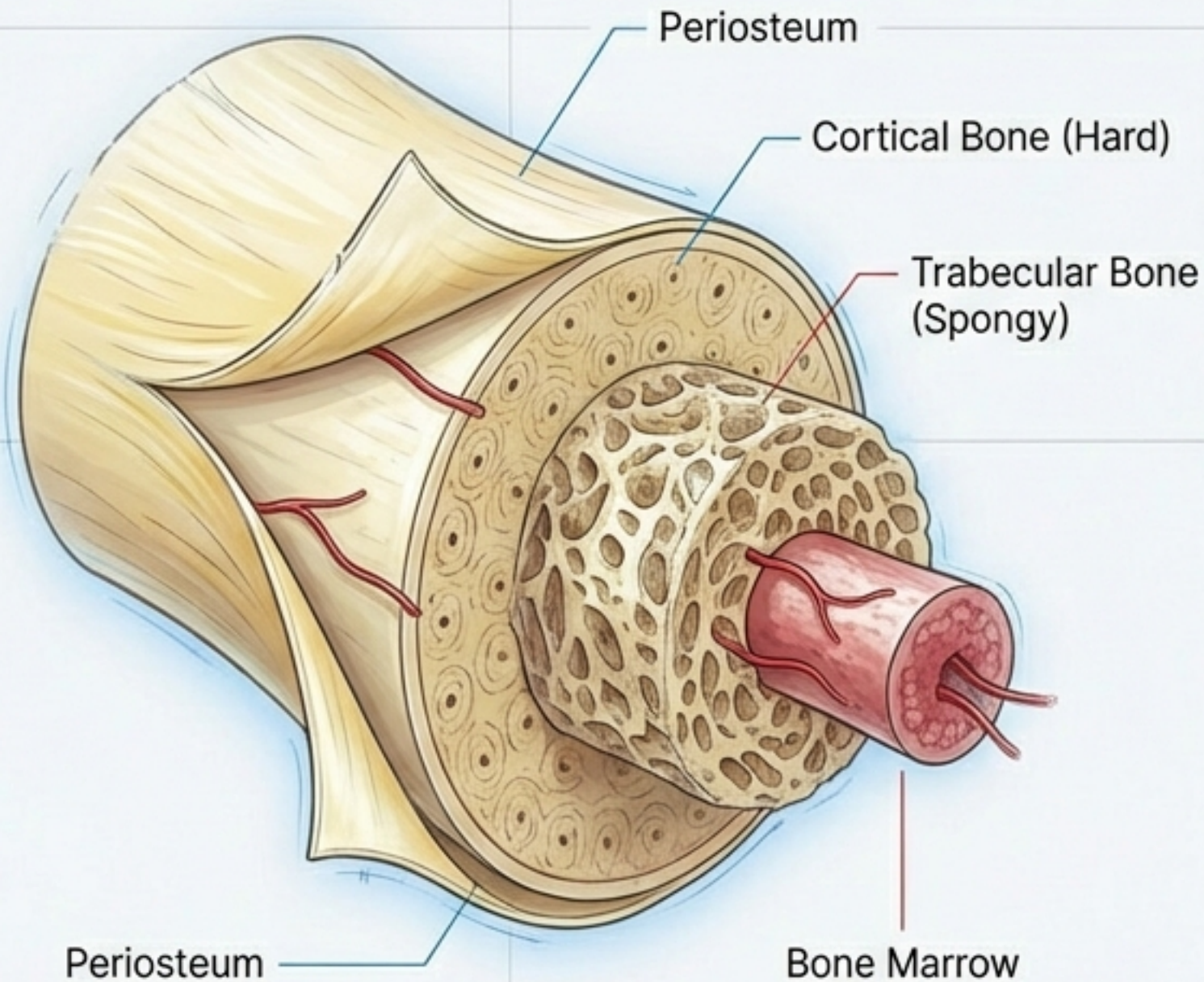
Complex shapes for specific purposes.

Sesamoid Bones (Friction Reduction)



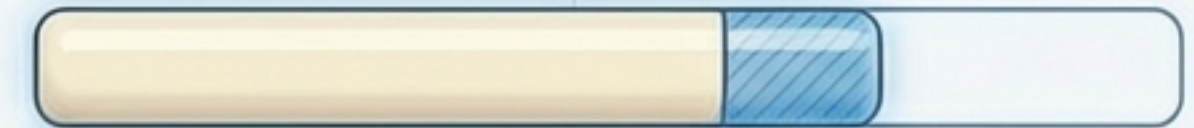
Change pressure & friction at a joint.

⊕ The Materials Science of Bone Tissue



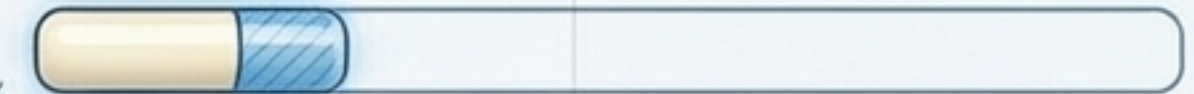
Chemical Composition:

%



Calcium Carbonate & Phosphate (60-70%)

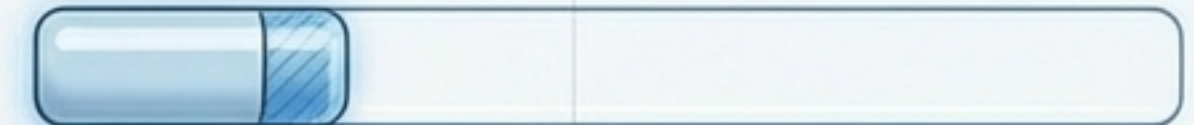
Provides stiffness and resistance to compressive forces.



Collagen Protein

Provides flexibility and resistance to tensile (pulling) forces.

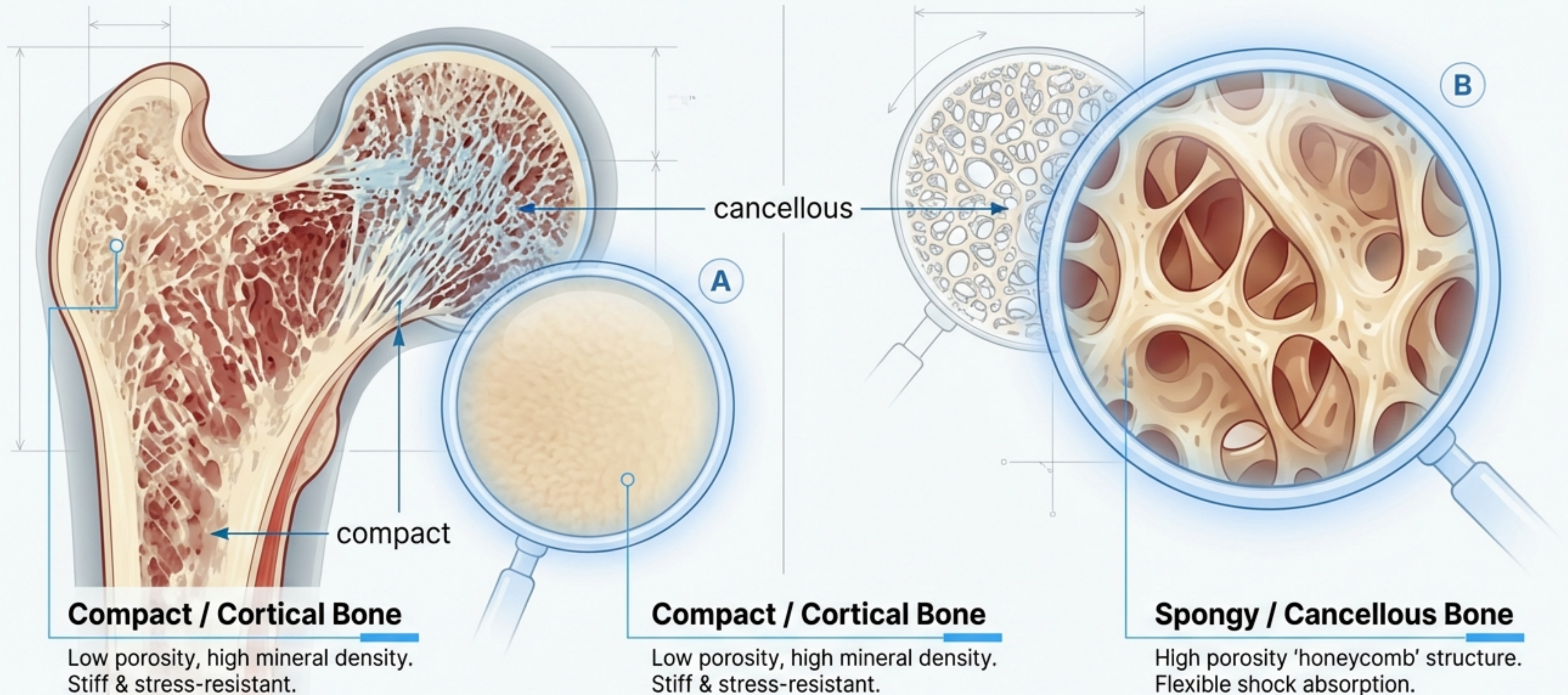
Note: Loss of collagen leads to brittleness.



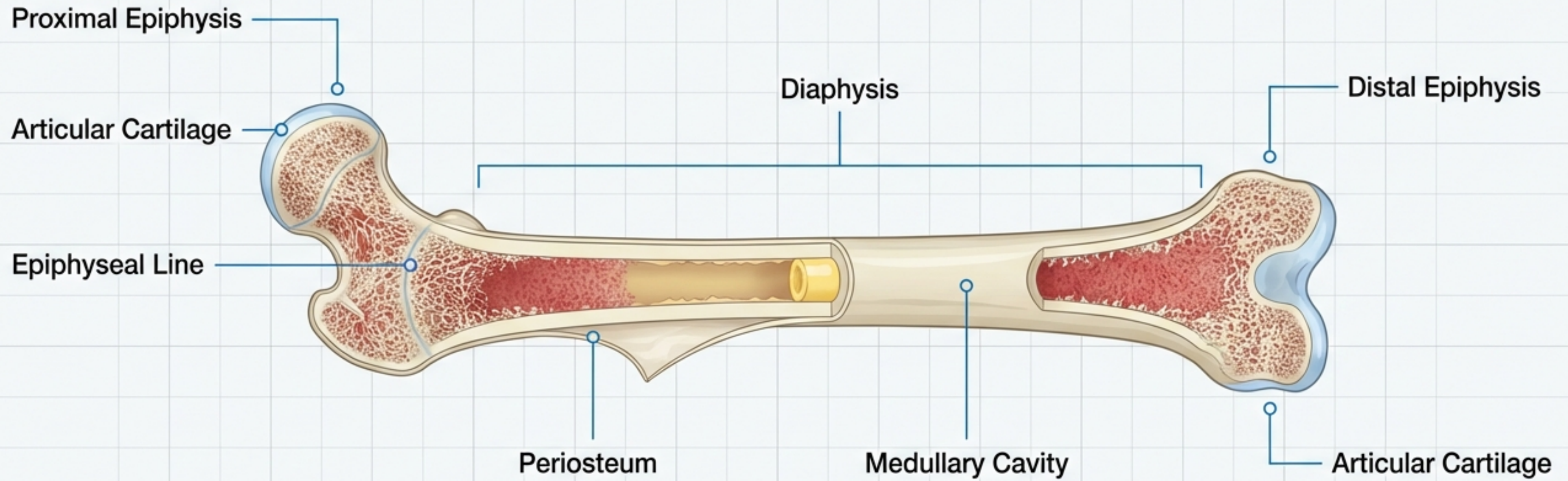
Water (20%)

Essential for bone health (vs 60% in soft tissue).

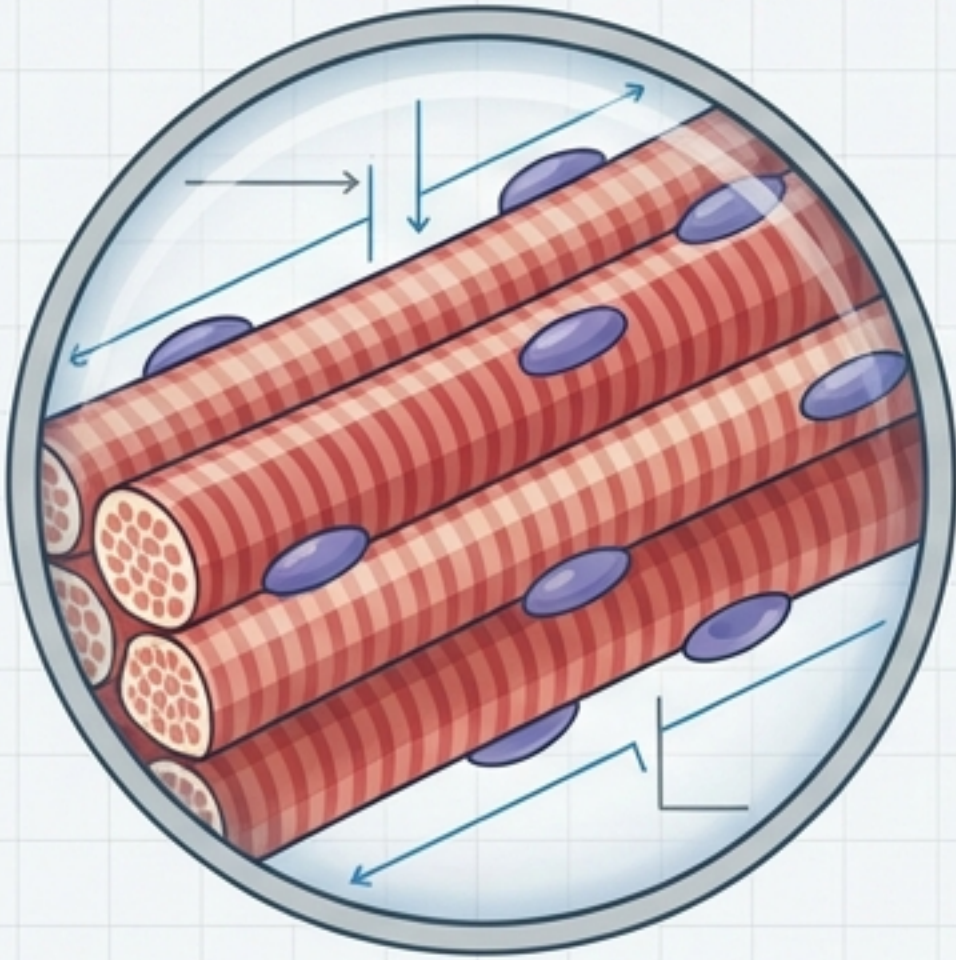
Structural Engineering: Cortical vs. Cancellous



Anatomy of a Long Bone

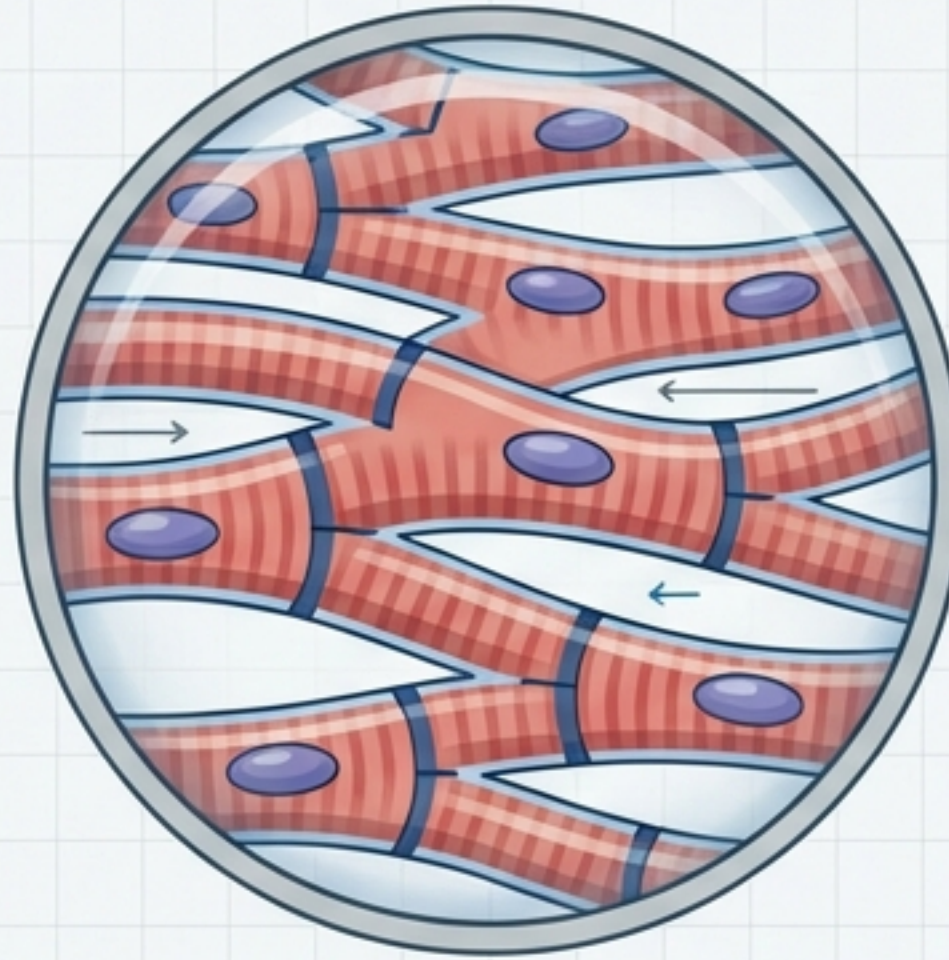


The Engine: Three Types of Muscle Tissue



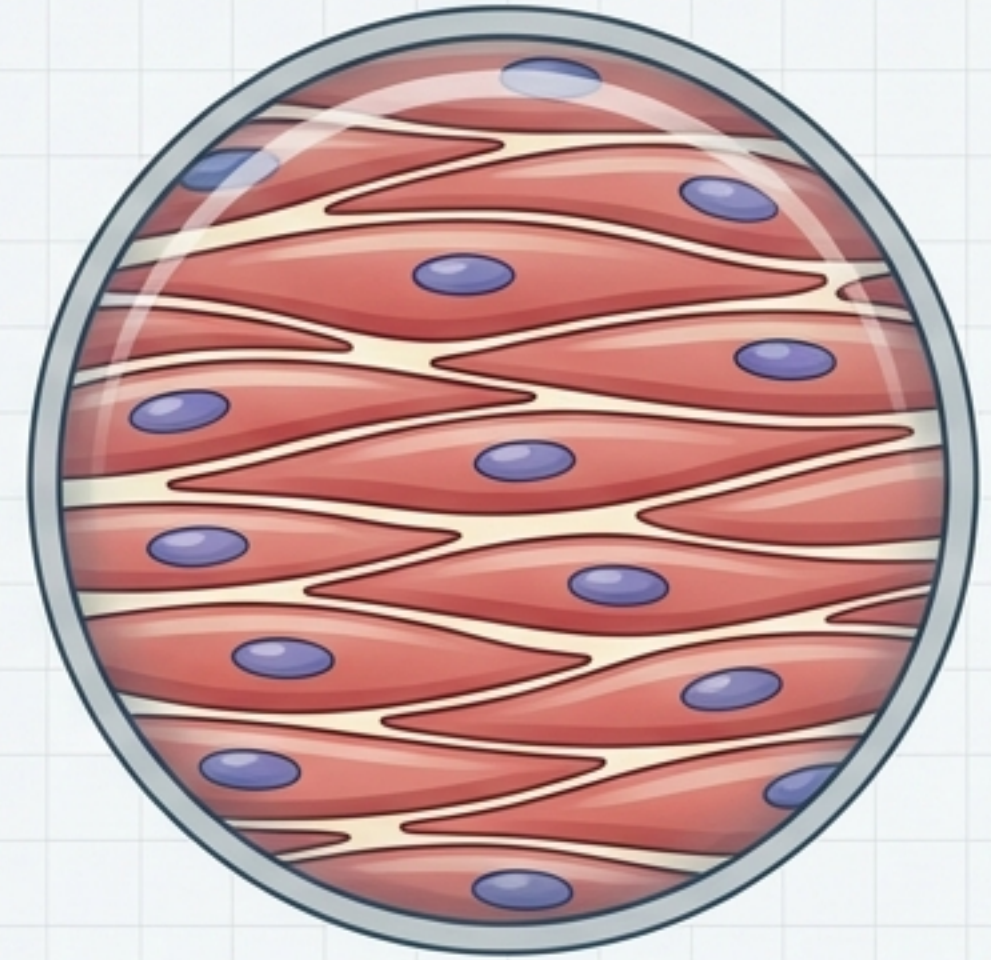
Skeletal Muscle

Attached to bone. Voluntary control via motor nerves. Contraction creates movement.



Cardiac Muscle

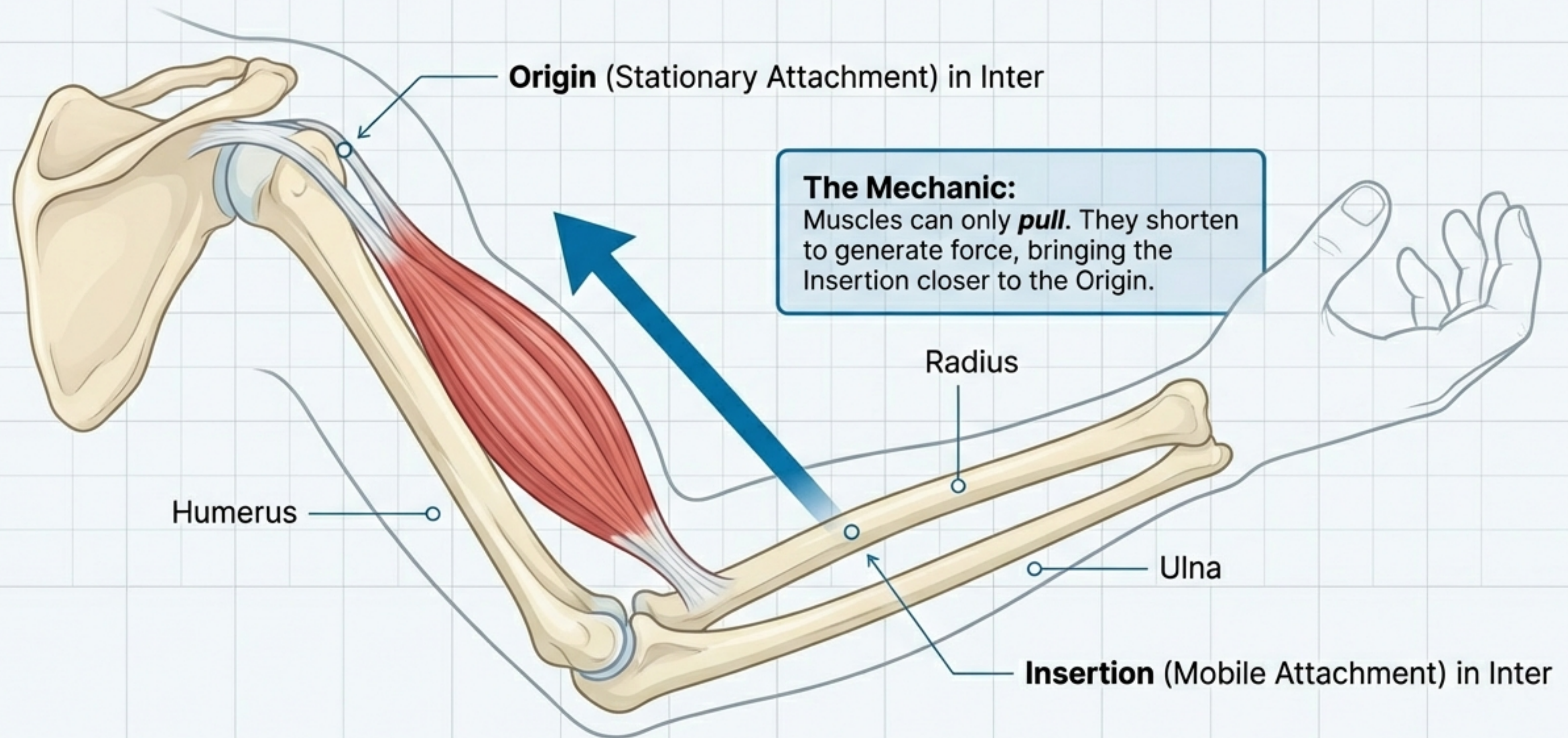
Heart only. Involuntary control. Fatigue-resistant pumping.



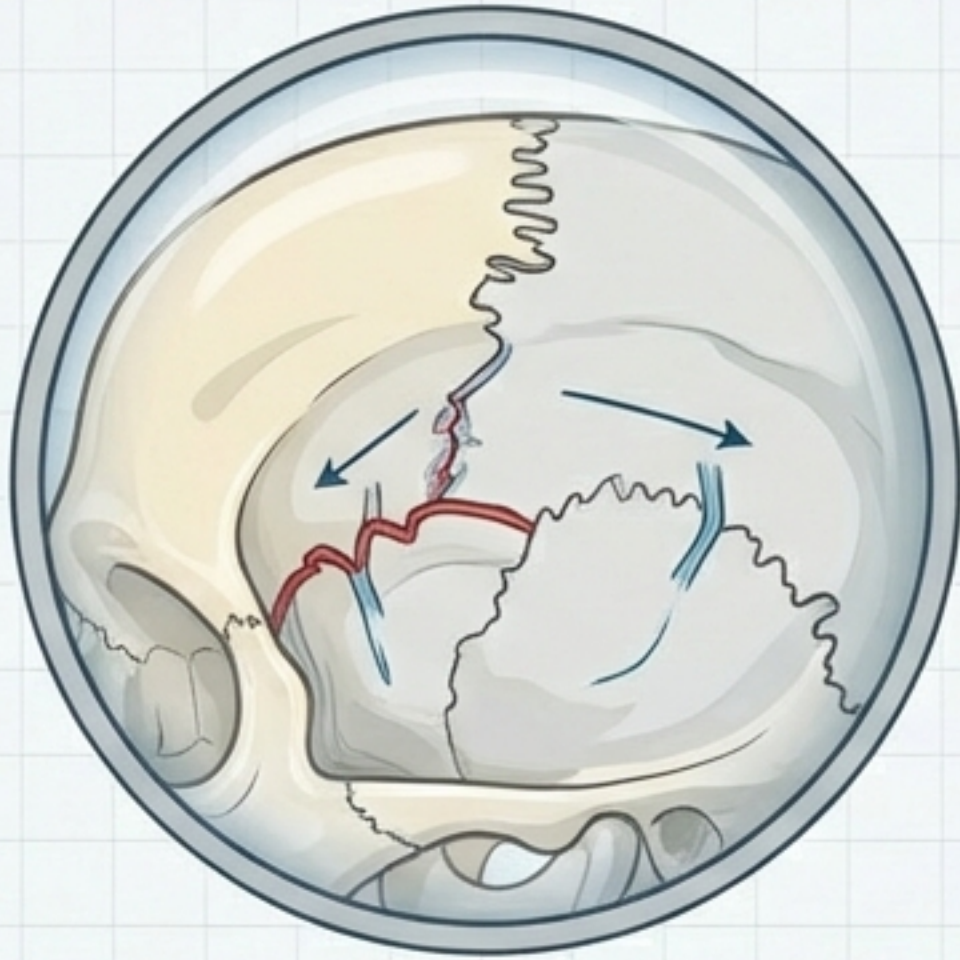
Smooth Muscle

Organs & vessels. Involuntary control. Slow, uniform contractions.

Skeletal Muscle Mechanics

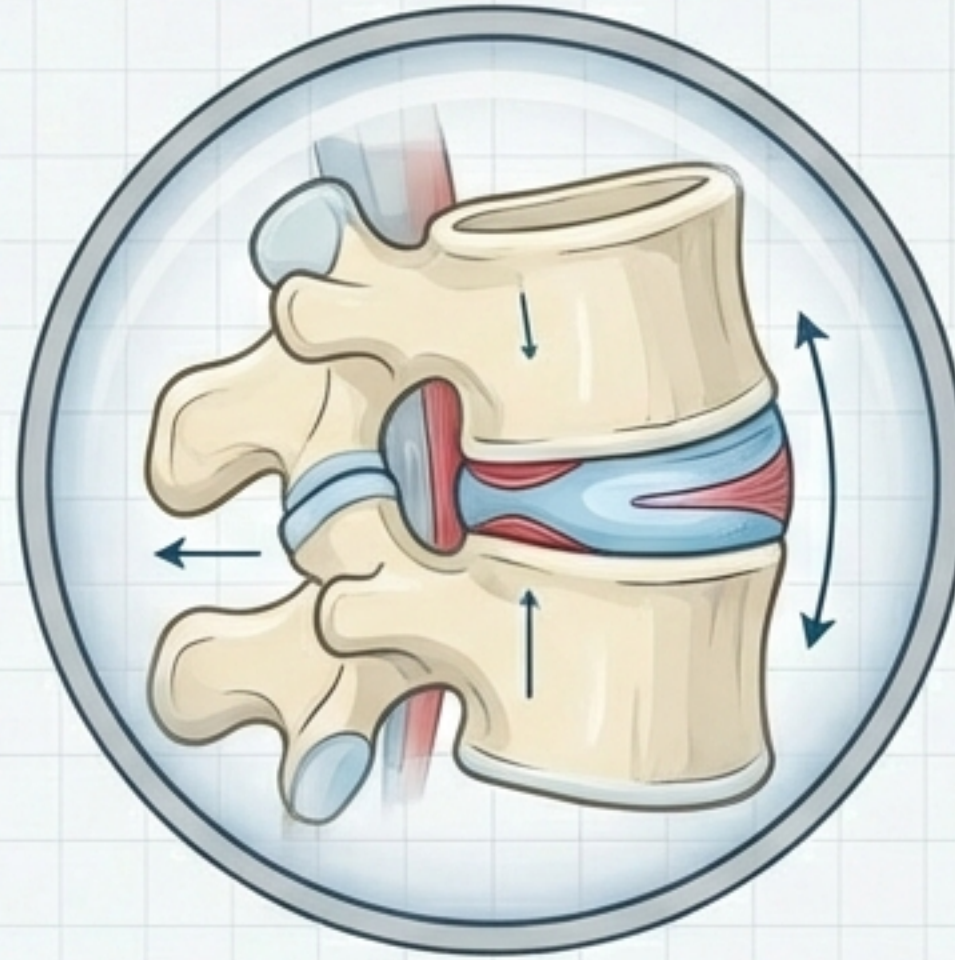


The Hinges: Classification of Joints



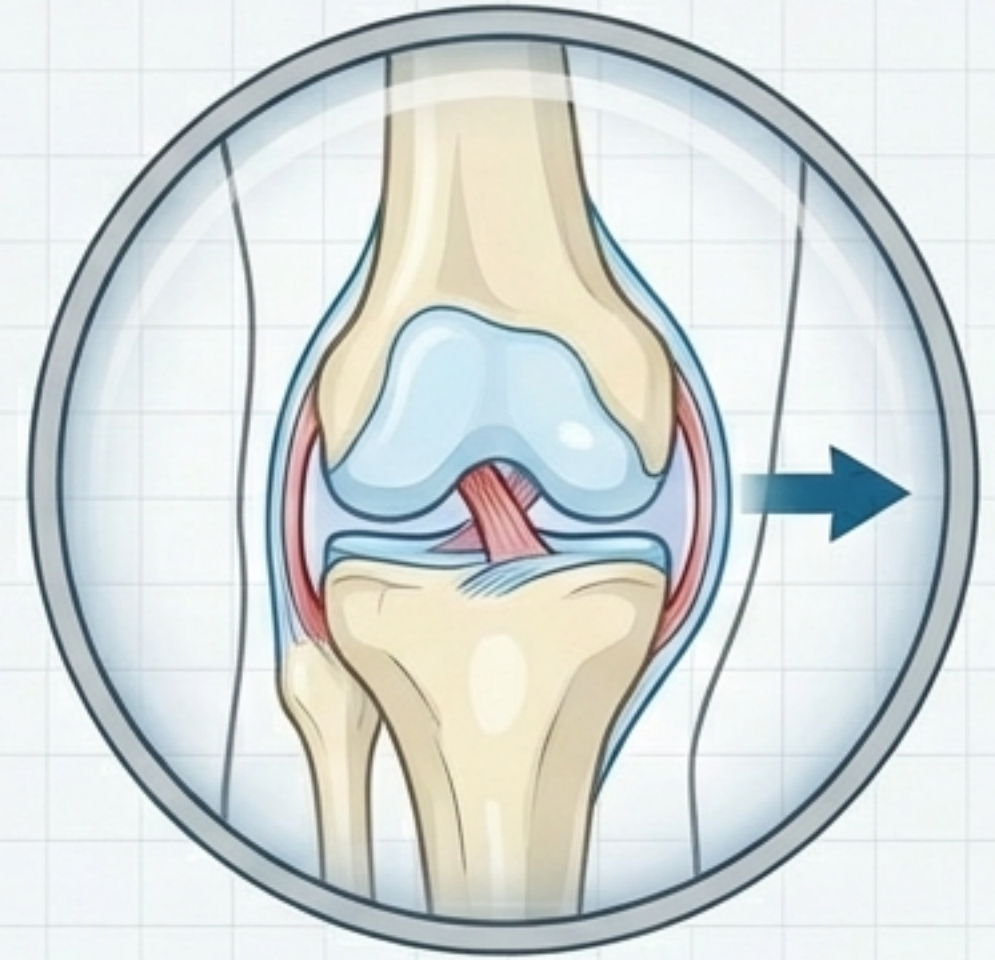
Fibrous Joints

No movement. Connected by connective tissue strands.



Cartilaginous Joints

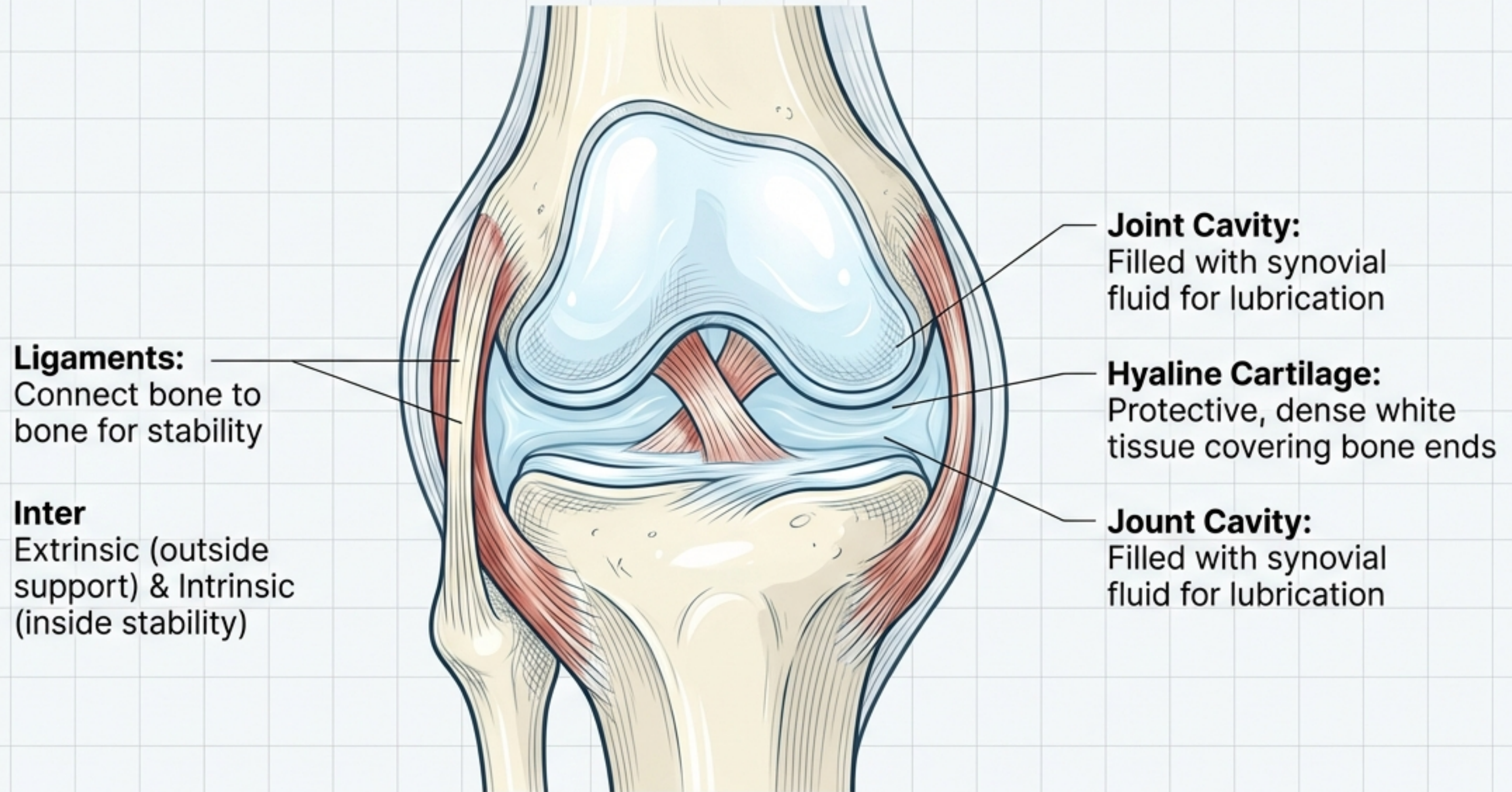
Limited movement. Shock absorbing connection.



Synovial Joints

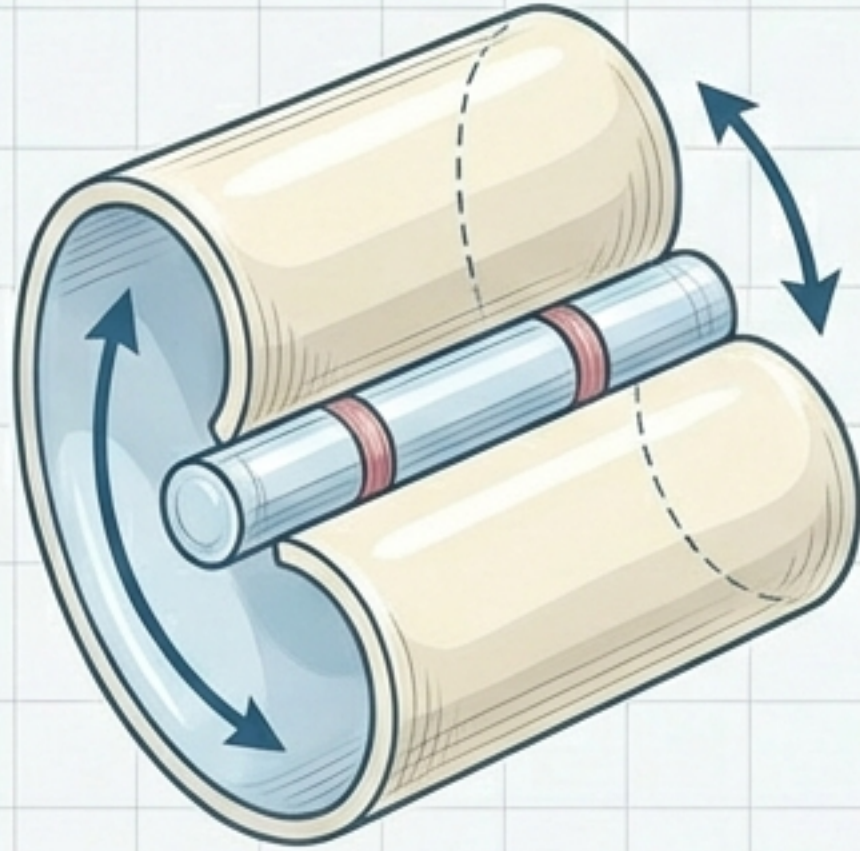
Greatest movement. Features a joint cavity. Most common in limbs.

Inside the Synovial Joint



Geometric Degrees of Freedom

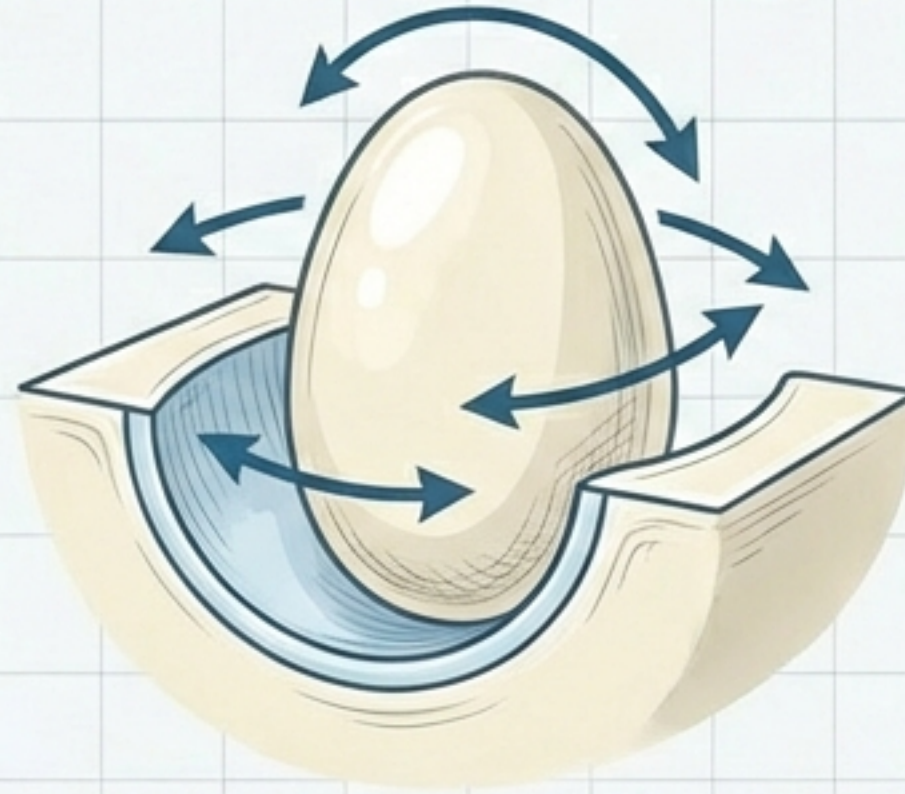
Uniaxial



Uniaxial

Movement around **one** axis.
(Rotation or Folding)

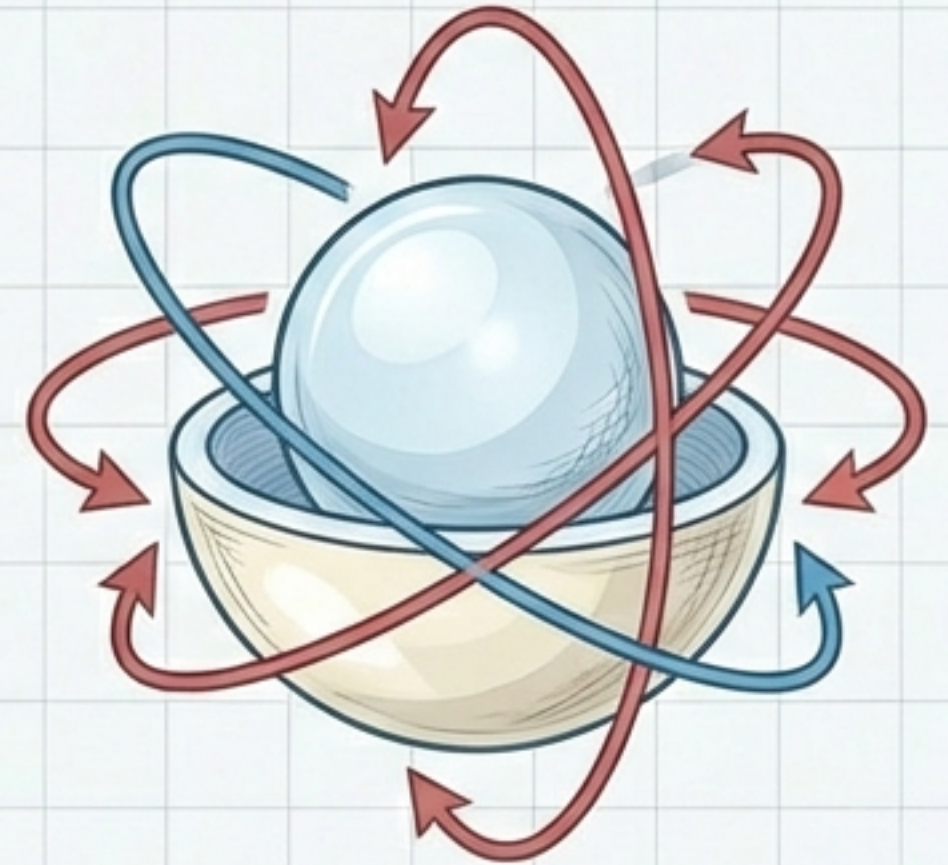
Biaxial



Biaxial

Movement around **two**
perpendicular axes.

Multiaxial

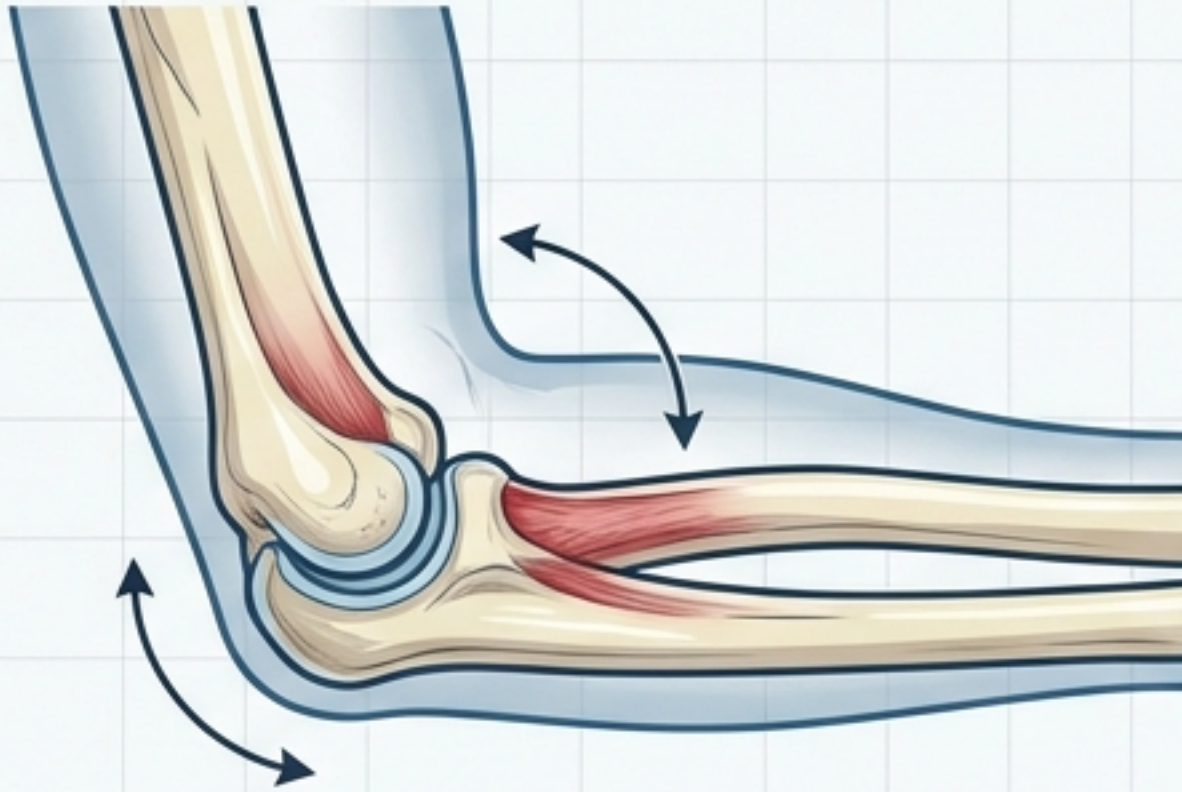


Multiaxial

Movement around
all three axes.

Uniaxial Joints: Precision in One Plane

Hinge Joint

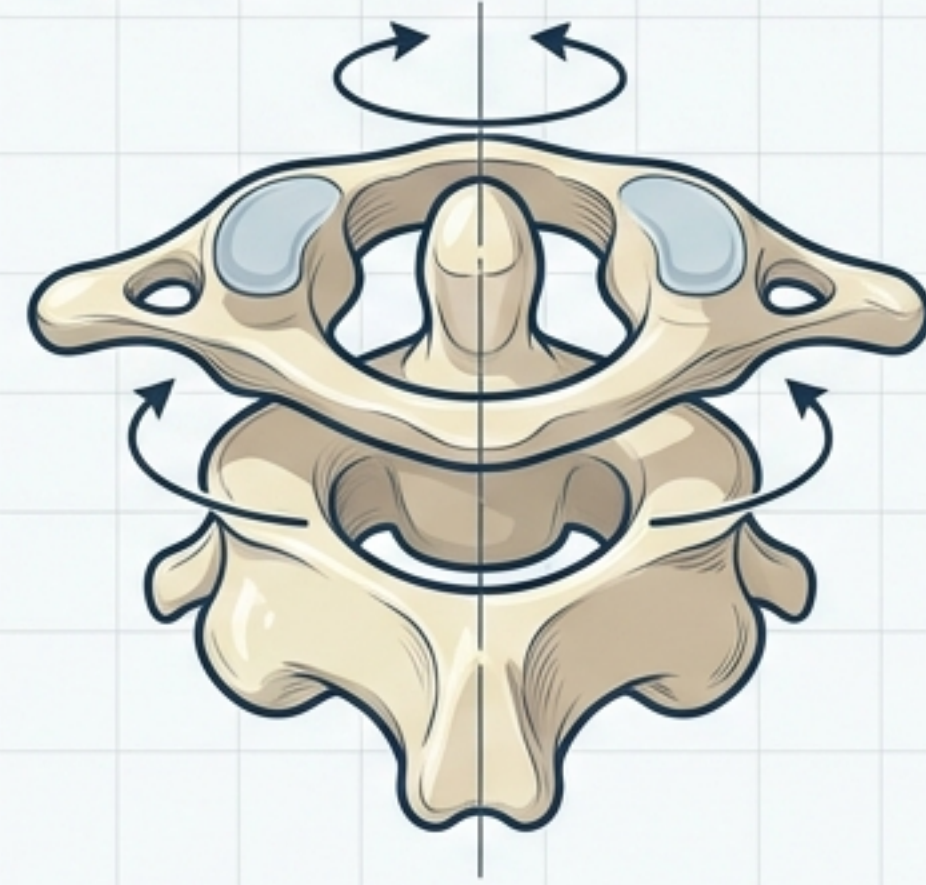


Convex and concave surfaces fit together.

Action: Flexion-Extension.

Example: Elbow.

Pivot Joint



One bone rotates around a single vertical axis.

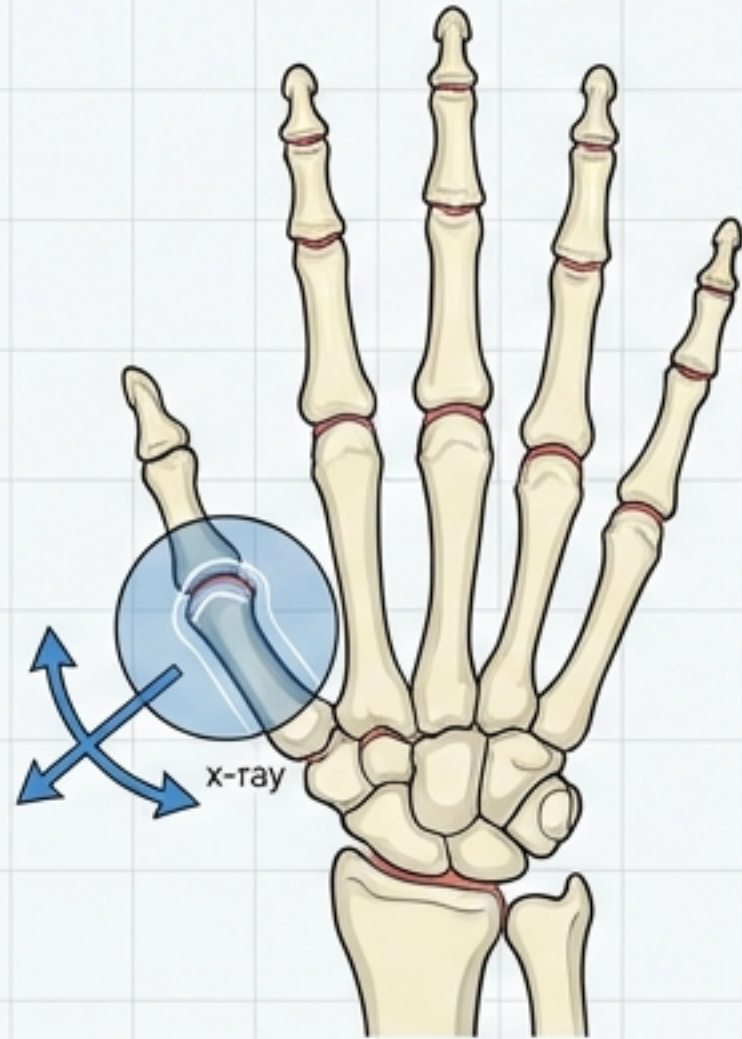
Action: Rotation.

Example: Neck (C1 & C2 Vertebrae).

****Gliding Joints:** Flat bone surfaces sliding (e.g., Acromio-clavicular).

Complex Articulation: Bi- and Multiaxial Joints

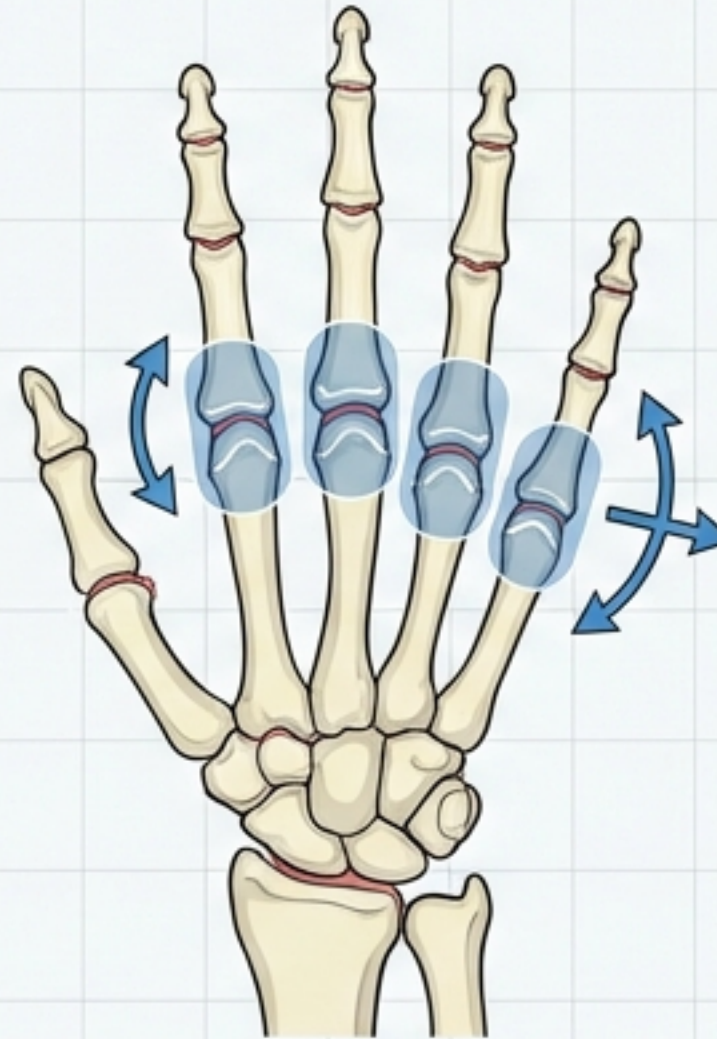
Saddle Joint (Biaxial)



Bones sit together like a rider on a saddle.

Example: Thumb (Carpometacarpal).

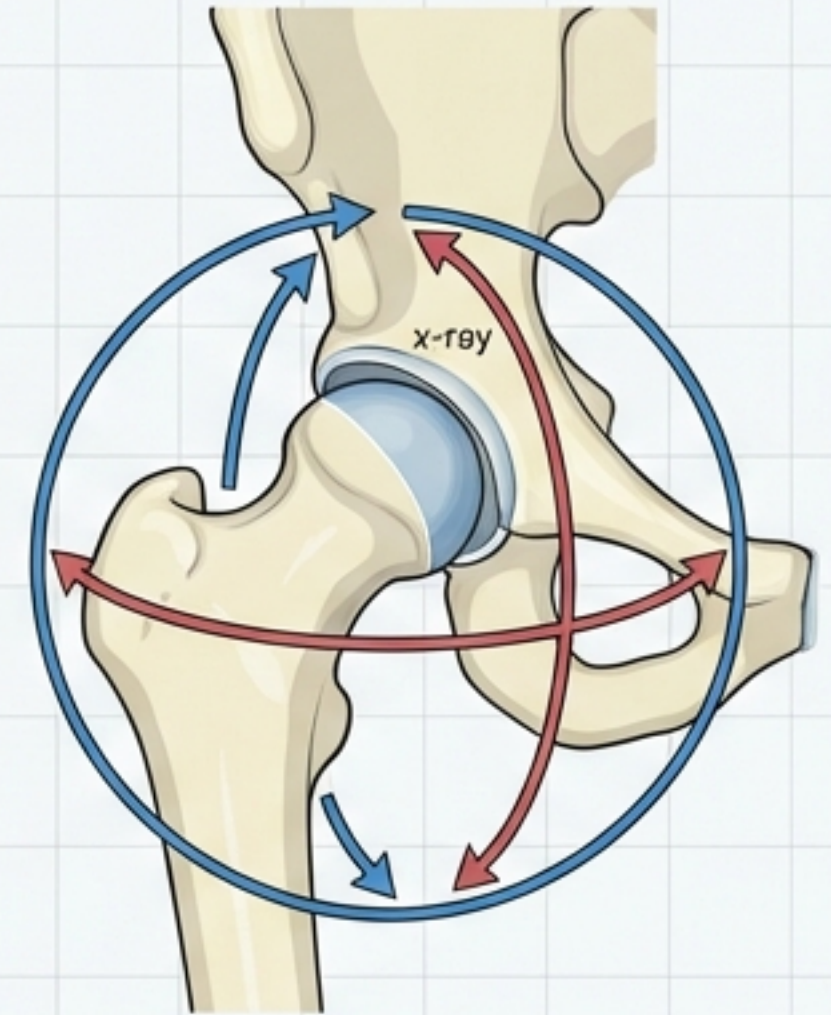
Condylloid Joint (Biaxial)



Ovular convex shape fits reciprocal concave surface.

Example: Knuckles (Metacarpophalangeal).

Ball & Socket (Multiaxial)



Rounded head fits into a cup-like receptacle. Maximum range of motion.

Example: Hip.

Synthesis: The Mechanics of Human Motion



The musculoskeletal system is a masterpiece of biological engineering.

1. **The Skeleton** provides the levers.
2. **The Joints** determine the range of motion.
3. **The Muscles** generate the force.

Result: The complex Kinesiology of life.